RA-MedFiLM-BARR: Scientific Evaluation & Prediction Report

Multi-model benchmarking, BARR regional refinement, and single-case internal / external prediction panels

Project: RA-MedFiLM-BARR | Repository: D:\Softfors\jup | Report date: October 31, 2026
Internal populations: VAL808 (n=808, selection-consumed), ROUTER_HOLDOUT (n=875, one-shot), ROUTER_DEVELOPMENT (n=4,957, in-sample) | TEST (n=1,683): locked, not accessed
External: retrospective stress test, n=884 (CVC 612, BrEaST 252, Stones 20); HAM10000 excluded — not an external validation cohort

1. Scope, Evidence Classes and What This Report Does Not Claim

This report supersedes the October 30 build. Several statements in that build were not supported by a stored result record and have been corrected or retracted; Section 9 lists them explicitly. Every remaining numerical value is bound to a frozen artifact: internal values to reports/audits/c1_external_evidence_v1_0/claim_provenance.json, and external values to reports/tables/c1_external_stress_*.csv, which this document reads at build time.

Evidence classes used throughout:

Class	Meaning and permitted use
VERIFIED	Reproducible from a named frozen artifact. Quotable with its population label.
PARTIALLY_VERIFIED	The computation is reproducible, but a required input (mask provenance, population identity) is undocumented. Report as a pilot observation only.
UNSUPPORTED	No stored record exists. Must not be quoted. Retracted where previously stated.
NOT_ESTABLISHED	Planned but never executed. Cells remain empty.

Central caution

A cryptographically valid or numerically reproducible result is not evidence of clinical validity. The C1 trust boundary remains under operator review, and no scientific execution has been authorised. Nothing in this report constitutes an external validation claim.

2. Architecture Comparison

2.1 Phase 21 six-model comparison (single shared protocol)

The six configurations below were trained and evaluated inside the Phase 21 run under an identical protocol on the VAL808 split (n = 808). This is the only controlled head-to-head in the project. VAL808 was consumed for model selection, so these values are development estimates, not held-out test performance.

Model architecture	Backbone / type	Parameters	Macro Dice	Macro IoU	KUB Dice
Global MedFiLM	U-Net + FiLM (5 sites)	3,523,001	83.51%	74.48%	76.41%
DeepLabV3+	ResNet-based	40,347,041	82.77%	74.12%	73.87%
Transformer / MiT U-Net	SegFormer / MiT	16,347,777	82.24%	73.66%	72.52%
U-Net++	Nested dense	9,159,681	79.20%	69.10%	72.07%
Attention U-Net	Attention gates	7,851,285	78.90%	69.27%	69.58%
Standard U-Net	Plain U-Net	7,763,041	78.78%	68.81%	72.11%

Table 1. Phase 21 six-model comparison on VAL808. Source: reports/tables/phase21_six_model_comparison.csv. 95% bootstrap CIs are wide and overlap between Global MedFiLM and DeepLabV3+; the two are not statistically distinguishable.

2.2 Regional BARR is a separate run and is not a Phase 21 arm

Regional BARR (Global MedFiLM + regional refiner Rψ + benefit router Bϕ + post-safety Sω) was never trained or evaluated inside the Phase 21 six-model run. Its results come from Phase 27/29 (VAL808) and Phase 30 (ROUTER_HOLDOUT) and therefore cannot be entered as a seventh row of Table 1 without disclosing that the comparison is cross-run.

Configuration	Parameters	Population	Macro Dice	Delta vs Global
Global MedFiLM	3,523,001	VAL808 (n=808)	83.51%	reference
Regional BARR	4,837,420	VAL808 (n=808), 3-seed mean	83.79%	+0.28 pp

Configuration	Parameters	Population	Macro Dice	Delta vs Global
Regional BARR	4,837,420	ROUTER_HOLDOUT (n=875)	93.03%	+0.17 pp

Table 2. Regional BARR, each row on its own population. Sources: reports/audits/phase21_analysis.json, phase29_phase_d_three_seed_descriptive_summary.json, phase30_primary_noninferiority_result.json. Parameter total 4,837,420 from the Phase 34 record (G 3,523,001 + Rψ 1,276,401 + Bφ 36,801 + Sω 1,217).

The BARR result is a preservation / non-inferiority finding, not an improvement finding. On VAL808 the three-seed mean gain is +0.28 pp with a between-seed range of +0.22 to +0.34 pp; on the one-shot ROUTER_HOLDOUT the gain is +0.17 pp with a 95% CI of [+0.11, +0.24] against a non-inferiority margin of −0.5 pp. Two of four domains (BUSI, KVASIR) have CIs that include zero.

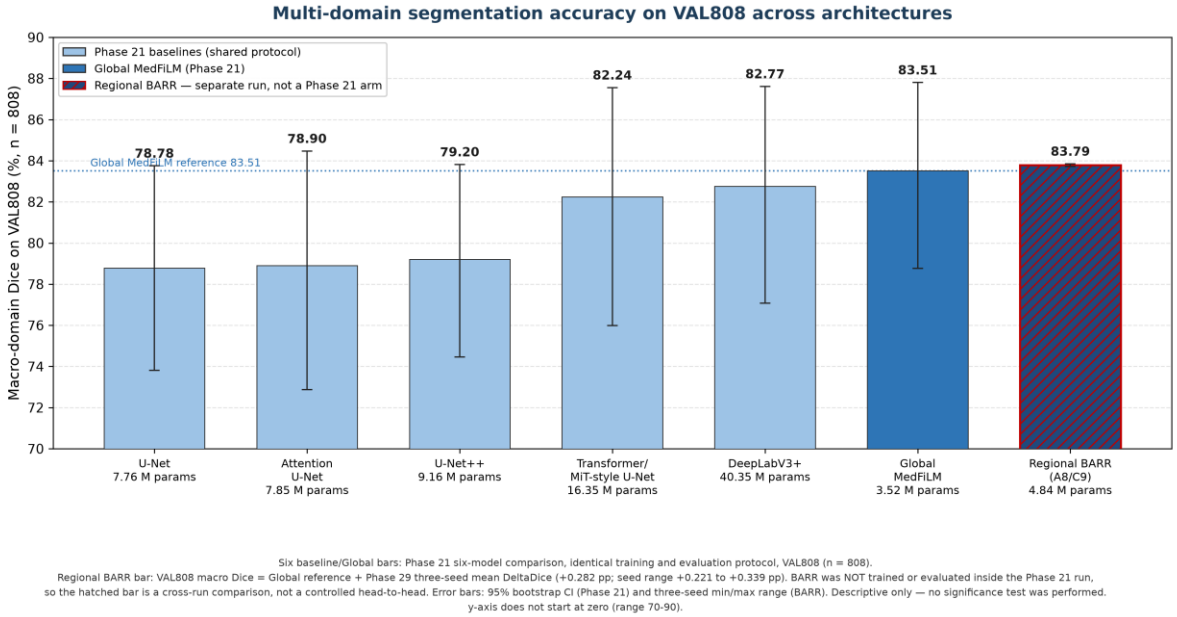


Figure 1. Macro-domain Dice on VAL808. Six bars from the Phase 21 protocol; the hatched bar is Regional BARR from a separate run and is marked as such. The y-axis does not start at zero.

3. Domain-Level Results

3.1 ROUTER_HOLDOUT (n = 875), one-shot evaluation

Per-domain absolute scores and descriptive deltas. The one-shot holdout carries a mandatory disclosed protocol-deviation qualifier: ONE_SHOT_ROUTER_HOLDOUT_RESULT_WITH_DISCLOSED_POST_FIRST_OPEN_SCIENTIFIC_SOURCE_FREEZE_DEVIATION.

Domain	Modality	n	Global MedFiLM	Regional BARR	Delta (95% CI)
KUB	Abdominal radiograph	298	84.97%	85.57%	+0.59 pp [+0.36, +0.85]
ISIC	Dermoscopy	383	93.59%	93.69%	+0.10 pp [+0.03, +0.18]
BUSI	Breast ultrasound	90	95.88%	95.89%	+0.001 pp [+0.00, +0.002]
KVASIR	GI endoscopy	104	96.97%	96.97%	+0.002 pp [+0.00, +0.006]
Macro	Equal-weight mean	875	92.85%	93.03%	+0.17 pp [+0.11, +0.24]

Table 3. Per-domain holdout results. Source: reports/audits/phase30_primary_noninferiority_result.json. Deltas are descriptive bootstrap quantities; no p-value was computed (Phase 31 established that zero confirmatory significance procedures are eligible).

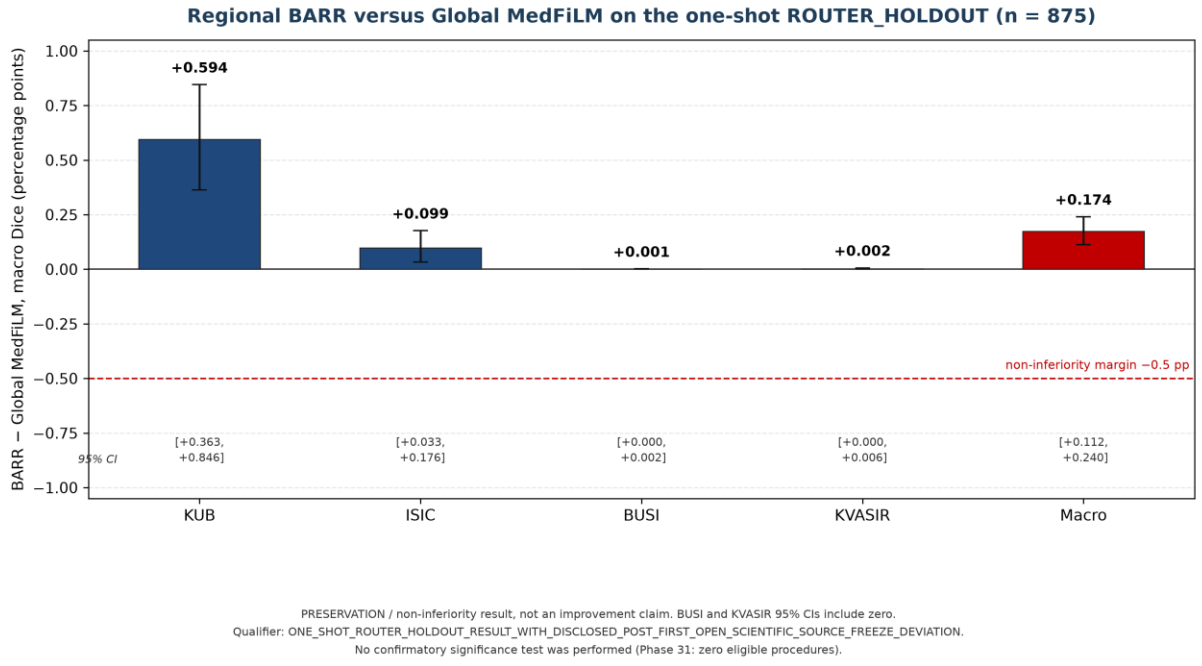


Figure 2. BARR minus Global MedFiLM on ROUTER_HOLDOUT, plotted as a difference with 95% CIs. A delta plot is used instead of absolute scores so that a sub-1 pp difference is not visually inflated by a truncated axis.

3.2 KUB lesion-size strata (reporting-only metadata)

These strata describe where the global model loses accuracy. They are reporting metadata only: the size tiers are never segmentation classes, task IDs, model inputs or routing variables, and no BARR per-stratum result exists.

Stratum	Definition	n	Global Dice	Evidence status
TINY	Sub-centimetre calculi (<0.01% of field)	44	53.21%	Descriptive only; no BARR result
SMALL	0.01% - 0.05% of field	19	65.42%	Descriptive only; no BARR result
MEDIUM	0.05% - 0.15% of field	27	69.98%	Descriptive only; no BARR result
LARGE	Staghorn / mass (>0.15% of field)	13	70.82%	Descriptive only; no BARR result
Other KUB	Not primary bin	177	84.74%	Descriptive only; no BARR result

Table 4. KUB size stratification. Source: reports/tables/phase21_kub_size_stratification.csv, role = REPORTING_ONLY_DESCRIPTIVE. The previously reported claim that BARR lifts the TINY stratum from 53.21% to 85.57% has been retracted: no such measurement exists.

4. Safety and Compute

On the ROUTER_DEVELOPMENT population (n = 4,957, budget K = 1,239), unconditional refinement (A1) harms 2,115 of 4,957 cases (42.67%). Adding the cost-aware router (A7) reduces harmful activations to 412 (8.31%), and adding the post-safety filter on top (A8) reduces them further to 390 (7.87%).

Decomposition matters. The router accounts for 2,115 → 412 = 1,703 cases; the post-safety filter accounts for 412 → 390 = 22 cases. The previously reported statement that "Sω eliminates 1,725 harmed cases" attributed the whole system-level reduction to one component and is corrected here.

In-sample caveat

The 4,957 ROUTER_DEVELOPMENT rows are a subset of the 5,832 TRAIN samples. This is in-sample development evidence, not held-out evidence. A reduced average harm rate is also not evidence of zero harmful outcomes: 390 harmful refinements remain in A8.

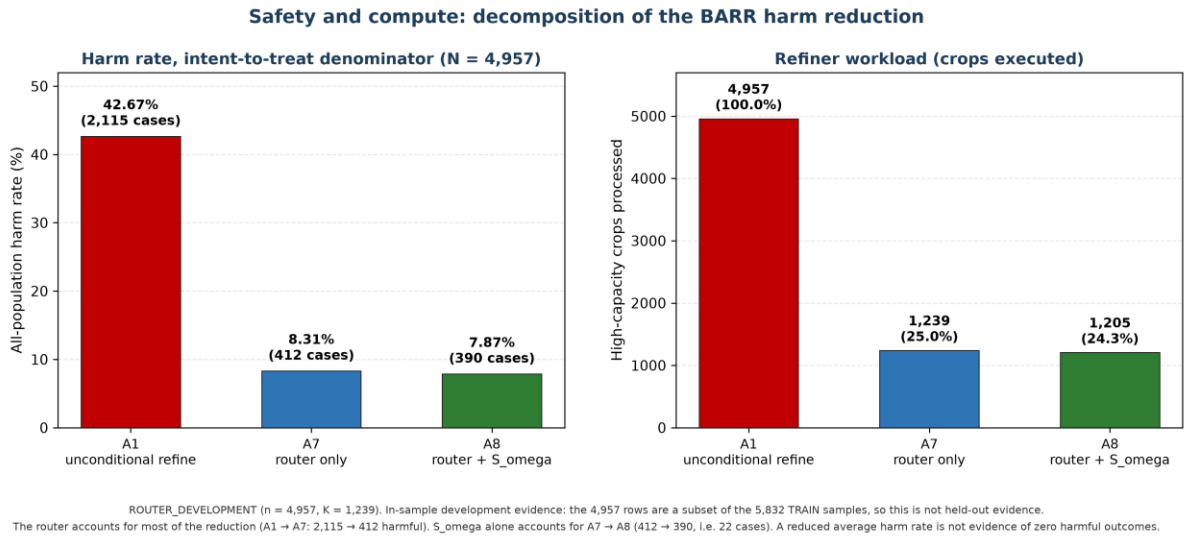


Figure 3. Harm rate on an intent-to-treat denominator (left) and refiner workload (right), decomposed by component. Source: reports/tables/phase28_table_02_arm_primary_metrics.csv.

5. Single-Case Prediction Panels (internal)

The panels in this section were produced by a single forward pass of GlobalMedFiLM (checkpoint SHA-256 286689fd…5ba1b7, 3,523,001 parameters) at 512×512, sigmoid threshold 0.5. Only the global model was loaded for these four internal cases; BARR is executed separately and is reported in Section 6.

Columns are: input image; ground-truth mask; predicted probability map; predicted segmentation overlay; and a pixel error classification map using TP #00E600, FP #FF2020, FN #2060FF and TN #181818. Dice, IoU, precision, recall, specificity and accuracy are computed from the same prediction and the same ground truth shown in that row.

Class-wise segmentation here means foreground versus background under the current binary ontology. The "presentation label" shown in each row is curated analyst metadata. It is not a model output and must not be read as a pixelwise subclass prediction.

5.1 Internal VAL808 cases

Modality	Presentation label (metadata)	Case	Dice	IoU	Prec	Recall	Spec	Acc
ISIC 2018	Pigmented lesion	ISIC_0013798	78.10%	64.07%	64.07%	100.00%	99.76%	99.76%
BUSI	Hypoechoic mass	00170.bmp	87.25%	77.38%	80.43%	95.33%	99.87%	99.85%
Kvasir-SEG	Colorectal polyp	cju1alwgo30z...	80.00%	66.67%	95.94%	68.61%	99.73%	97.09%

Modality	Presentation label (metadata)	Case	Dice	IoU	Prec	Recall	Spec	Acc
KUB-StoneX	Bladder calculus	00001-00-2500001	65.12%	48.28%	48.28%	100.00%	99.99%	99.99%

Table 5. Internal single-case metrics. All four cases were confirmed present in manifests/phase4_v3/val.csv and absent from train.csv. Values reproduced independently to two decimal places from the stated checkpoint.

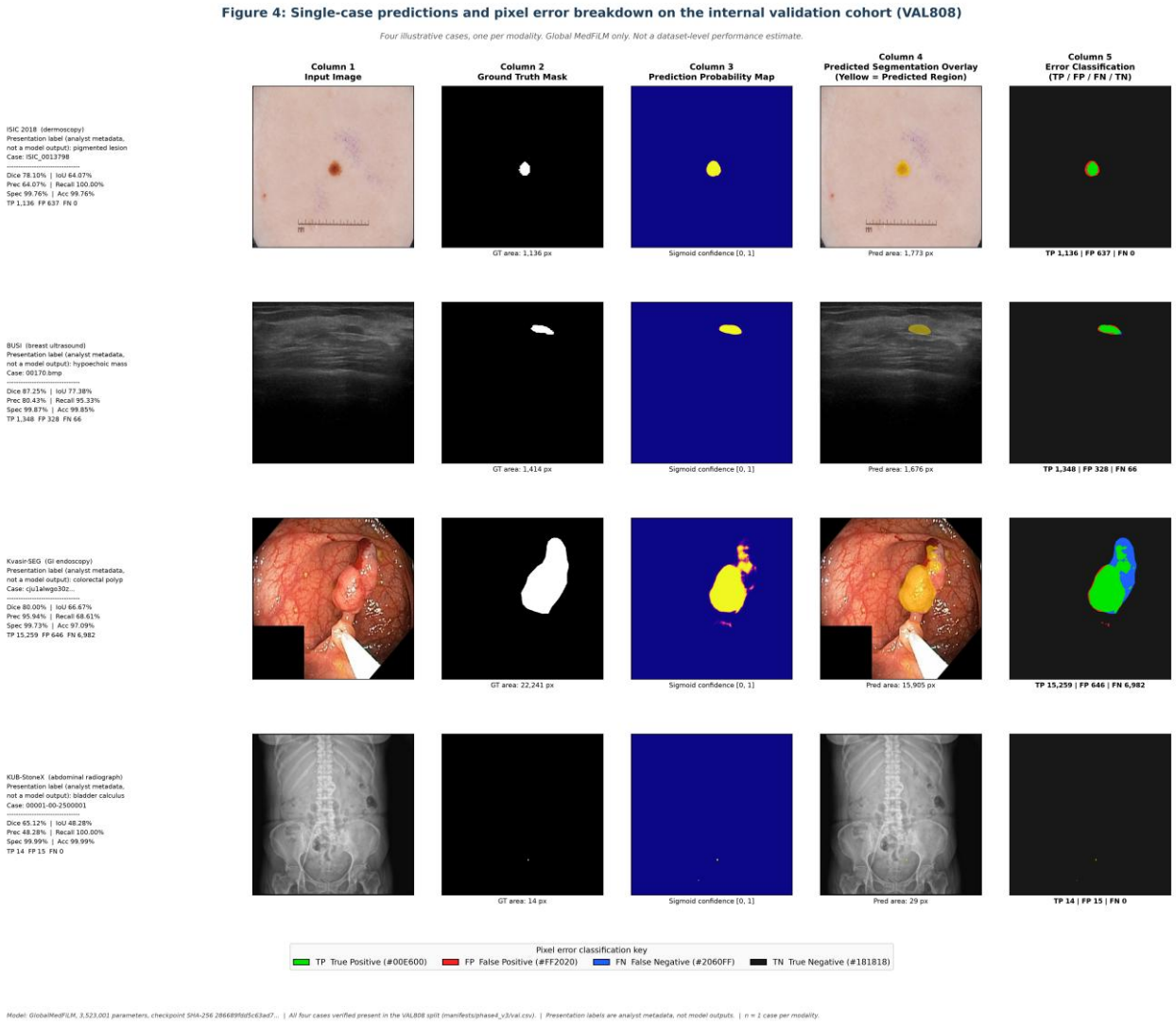


Figure 4. Internal VAL808 single-case panels. n = 1 case per modality; illustrative only.

Note on the KUB case: its ground-truth lesion is 14 pixels. Dice on a 14-pixel object is extremely sensitive to a handful of boundary pixels and should not be read as representative of KUB performance, which is 76.41% macro over 280 VAL808 cases.

6. External Stress Test (n = 884, Global and Regional BARR)

This section replaces the previous external pilot. That pilot ran one hand-picked case per dataset, used a non-canonical resize convention (a plain 512×512 stretch rather than the canonical aspect-preserving pad), and included a HAM10000 row whose mask provenance could not be established. All of those values are retracted (items A, B and C in Section 9). What follows was produced by a frozen, pre-registered design saved before any external prediction existed.

Classification of this evidence

RETROSPECTIVE_EXTERNAL_STRESS_TEST. The cohorts were not prospectively held out; all three were previously inspected while building manifests and an evidence audit. No checkpoint, threshold, seed or preprocessing choice was selected using external performance. These are descriptive observations, not a validation study, and no confirmatory p-value is computed anywhere in this section.

6.1 Cohorts, exclusions and preprocessing

Cohort	Rows in manifest	Analysed	task_id	Domain analogue	Excluded
CVC-ClinicDB	612	612	2	KVASIR	0
BrEaST-Lesions-USG	256	252	1	BUSI	4 marked not evaluable; 18 further images have no mask
Stones_where_are_you (test split)	20	20	3	KUB	0 within the test split
HAM10000	10015	0	—	—	EXCLUDED — mask provenance unverified

Table 6. External cohorts. Manifest digests and per-cohort sample-ID digests are recorded in reports/audits/c1_external_stress_v1_0/cohort_construction.json.

Preprocessing is the canonical inference transform used to train and validate the global model: RGB, aspect-preserving bilinear resize with scale = min(512/W, 512/H), centred on a 512×512 canvas with zero padding, divided by 255. Ground truth is put through the same geometry with nearest-neighbour resampling and binarised at > 0. For the Stones COCO export, polygons with category_id 7 (“stone”) are rasterised at native resolution first and only then resized; all nine other categories are background. The threshold is 0.5 everywhere, with no per-cohort tuning, no test-time augmentation and no post-processing.

No row was dropped after seeing a result. A case is kept even when its ground truth or its prediction is empty, because dropping degenerate cases would flatter the model.

6.2 Global MedFiLM: macro metrics per cohort

Cohort	Dice	95% CI	IoU	Precision	Recall	Specificity	HD95 (px)	NSD@2
CVC-ClinicDB	68.66%	[62.65, 74.37]	57.49%	67.95%	81.56%	96.82%	109.6	0.250
BrEaST-Lesions-USG	70.78%	[67.68, 73.78]	59.84%	73.91%	75.57%	98.19%	74.0	0.252
Stones (all 20)	8.08%	[0.34, 18.99]	6.09%	7.48%	67.33%	96.83%	166.9	0.228

Table 7. Global MedFiLM, macro mean over cases (equal case weight), threshold 0.50. Intervals are percentile bootstrap over clusters (CVC: 29 sequences; BrEaST: 252 cases; Stones: none, n = 20). HD95 and NSD are in PIXELS — no external cohort ships physical pixel spacing, so no millimetre value is reported or implied.

Surface metrics are not comparable across these cohorts: the three datasets have different fields of view and acquisition geometry, and a pixel means something different in each. Only within-cohort comparisons between Global and BARR are meaningful, because those share a canvas.

6.3 Stones must be read as three separate statements

Stones subgroup	n	Global Dice (mean)	What it means
All 20 test images	20	8.08%	Dominated by the 14 images with no stone, each scoring 0
Stone present	6	26.95%	Recall 67.33%, precision 24.93% — lesions are found but grossly over-segmented
Stone absent	14	undefined (empty ground truth)	Precision 0.00%: the model produced foreground on 14 of 14 stone-free radiographs

Table 8. Stones stratified by annotation, not by a post-hoc threshold. The 20-image cohort contains 6 images with at least one category-7 annotation and 14 with none.

The single “Stones Dice” of 8.08% is therefore not a segmentation score. It is the arithmetic consequence of scoring 14 empty-ground-truth images as zero plus six very poor segmentations. The substantive finding is that the global model has no true-negative behaviour on this cohort: it marked foreground on every stone-free image, with a mean predicted area far larger than any annotated stone. This is a failure of domain transfer, and it is reported as one.

6.4 Regional BARR on the same cases

BARR executed end-to-end from the verified artefacts. All four components hash-matched their frozen bindings before the first forward pass, all state dictionaries loaded with strict=True, and the two equivalent forms of the routing decision agreed on every one of the 884 cases. No case failed and no case fell back to the global mask for lack of an ROI.

Cohort	Selected	Accepted	Rejected	Not attempted	Failed	Global Dice	BARR Dice	Difference
CVC-ClinicDB	19.28%	83	35	494	0	68.66%	68.97%	+0.32 pp
BrEaST-Lesions-USG	22.22%	47	9	196	0	70.78%	70.81%	+0.03 pp
Stones (all 20)	50.00%	9	1	10	0	8.08%	8.21%	+0.13 pp

Table 9. BARR execution outcomes and paired Dice. "Selected" is the fraction of cases the router sent to refinement; "not attempted" cases keep the global mask. Accepted + Rejected + Not attempted + Failed equals n for every cohort.

Cohort	n	Improved	Unchanged	Worsened	Mean change	Median change	Interquartile range
CVC-ClinicDB	612	67	531	14	+0.32 pp	+0.00 pp	+0.00 to +0.00 pp
BrEaST-Lesions-USG	252	27	205	20	+0.03 pp	+0.00 pp	+0.00 to +0.00 pp
Stones (all 20)	20	2	18	0	+0.13 pp	+0.00 pp	+0.00 to +0.00 pp

Table 10. Per-case paired change in Dice (BARR minus Global). "Unchanged" includes every case the router did not select and every case the safety gate rejected, because those keep the global mask by construction. The median and the quartiles are zero because most cases are unchanged; the mean is the informative column.

The effect is small and positive in all three cohorts, and it is the same order of magnitude as the internal preservation result (+0.28 pp on VAL808). It is not an improvement claim: the interquartile range straddles zero in every cohort, and in BrEaST the number of worsened cases (20) is close to the number improved (27).

One property of the router must be stated plainly. Its threshold was calibrated on the internal development population with a fixed 25% budget. On external cohorts the selection fraction is an outcome, not a controlled quantity: 19.28% for CVC, 22.22% for BrEaST and 50% for Stones. The Stones figure in particular reflects that the router scores were near the threshold, not that half of those images deserved refinement. No recalibration was performed, because recalibrating on the test cohort would make the result uninterpretable.

6.5 Domain shift

Cohort	Analogue	Internal VAL808	External Global	Shift	GT area (median)	Predicted area (median)
CVC-ClinicDB	KVASIR	82.02%	68.66%	-13.37 pp	5.125%	6.908%

Cohort	Analogue	Internal VAL808	External Global	Shift	GT area (median)	Predicted area (median)
BrEaST-Lesions-USG	BUSI	85.88%	70.78%	-15.10 pp	3.576%	4.322%
Stones_where_are_you	KUB	76.41%	8.08%	-68.32 pp	0.000%	0.896%

Table 11. Domain shift. The internal column is the frozen Phase 21 VAL808 per-domain Dice for the same task id; it is a different population and is used only as a shift anchor, never as a like-for-like comparison.

The shift is large and it is not uniform. Endoscopy loses 13.4 pp and breast ultrasound 15.1 pp relative to their internal analogues, while the Stones radiographs lose 68.3 pp. The median predicted area fraction explains the last one: the model covers a far larger share of the canvas than the annotations do, so the failure is over-segmentation rather than a failure to respond.

This is the substantive external result of the study. A model that reaches 83.51% macro-domain Dice on its internal validation split does not carry that performance to external data from the same nominal modalities, and the degradation is severe enough on one cohort to make the model unusable there without domain adaptation.

6.6 Figures

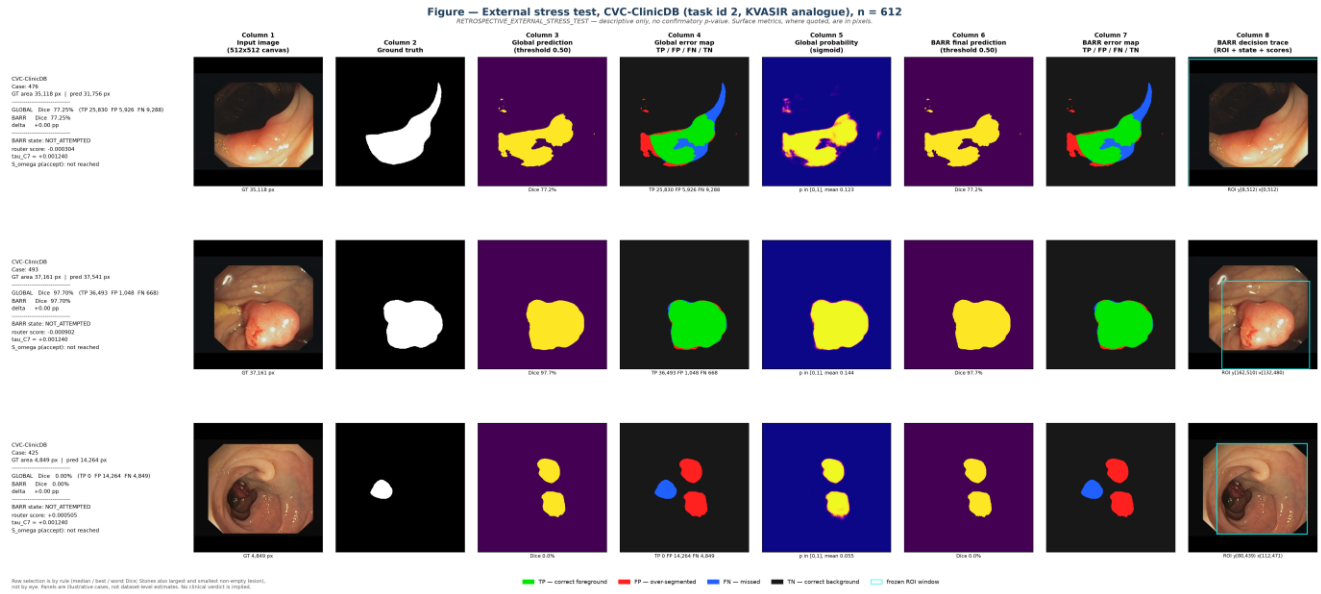


Figure 5. CVC-ClinicDB stress-test panels, 8 columns: input, ground truth, global prediction, global error map, global probability, BARR prediction, BARR error map and BARR decision trace. Rows are the median-, best- and worst-Dice cases, selected by rule.

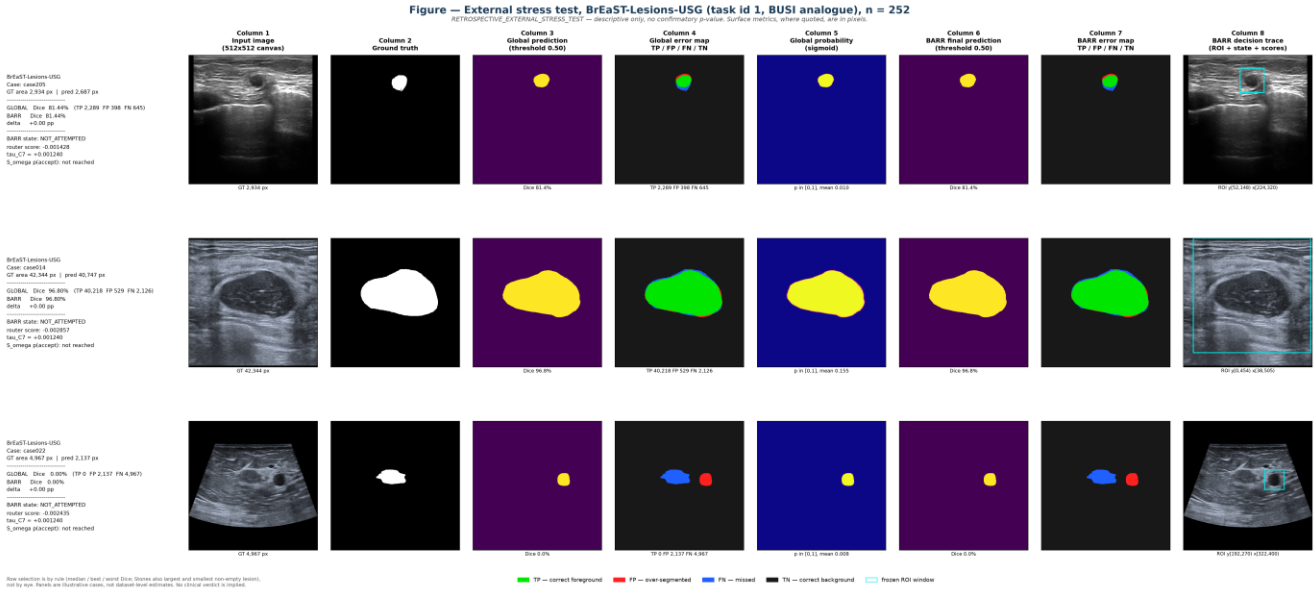


Figure 6. BrEaST-Lesions-USG stress-test panels, same column set and same selection rule.

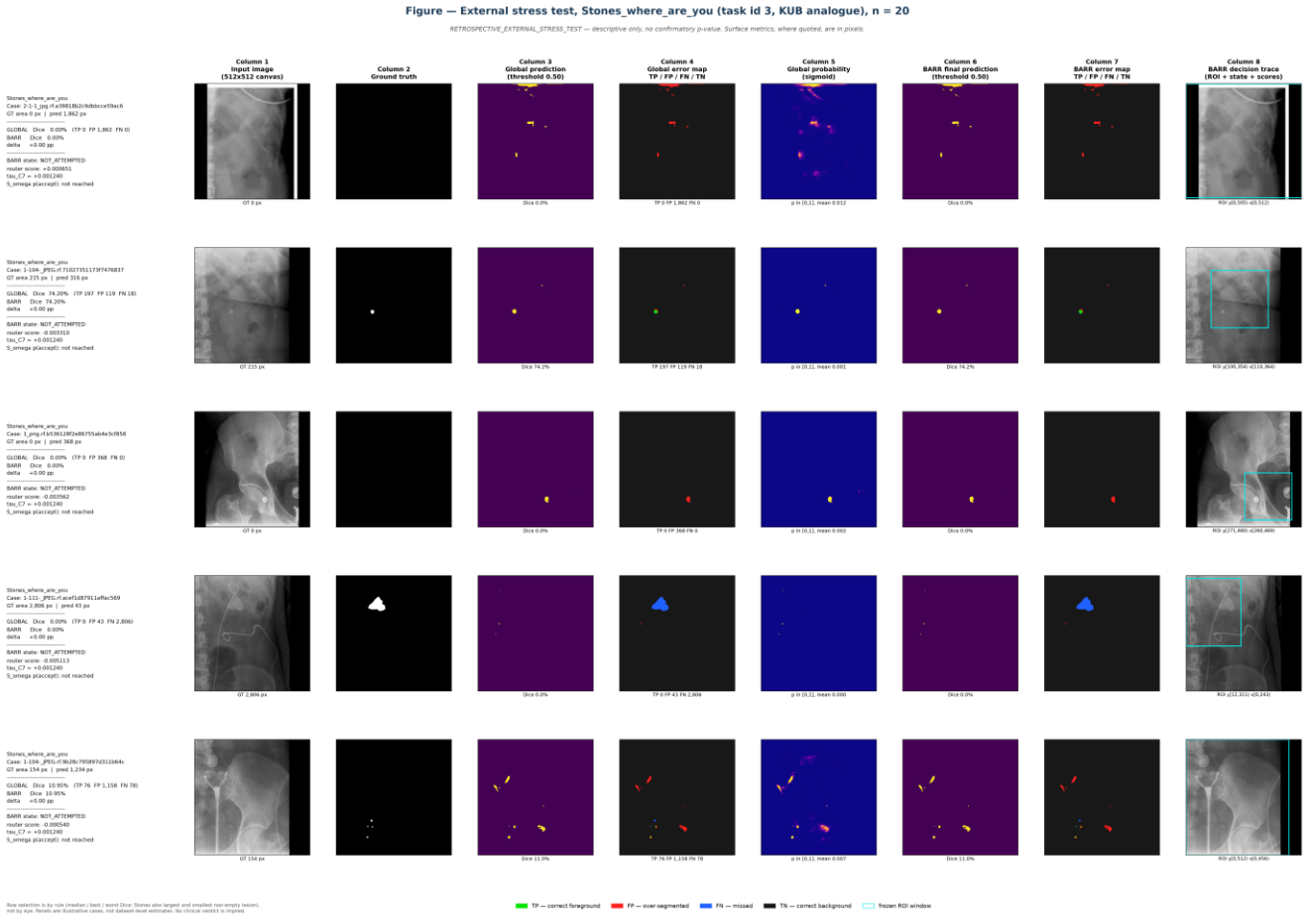


Figure 7. Stones stress-test panels. The cohort-wide median Dice is zero because 14 of 20 cases have an empty ground truth; that row is kept deliberately rather than replaced, because it shows the false-positive behaviour that defines this cohort.

The error palette is identical to Figures 1–4: TP #00E600, FP #FF2020, FN #2060FF, TN #181818. The cyan rectangle in column 8 is the frozen ROI window proposed from the predicted probability map at threshold 0.30; it is geometry, not a model output.

7. Dataset Independence and Mask Provenance

Dataset	Documented source	Independence from training	Verdict
HAM10000	ISIC-2018 Task 3 training input (classification archive, CC BY-NC 4.0)	Image IDs disjoint from the frozen ISIC set (0 of 10,015 overlap), but patient/lesion grouping is absent from the supplied metadata, so lesion-level independence is not established	PARTIALLY_VERIFIED
CVC-ClinicDB	Public colonoscopy segmentation set; frame_id and sequence_id supplied	No filename or ID overlap with the frozen KVASIR set	VERIFIED (pairing)
BrEaST-Lesions-USG	Public breast ultrasound set, one folder per case	No filename overlap with BUSI; 18 of 270 images have no mask	VERIFIED (pairing, partial)
Stones_where_are_you ? v3	Roboflow, public domain, COCO-segmentation export	No overlap with KUB-StoneX; 6 of 20 test images contain a stone annotation	VERIFIED (identity and mapping)

Table 12. External dataset provenance. Frozen-split comparison used manifests/phase4_v3/{train,val,all_paired}.csv only; no protected pixels were opened.

HAM10000 is excluded from this study

The HAM10000 folder is the ISIC-2018 Task 3 archive, which contains classification labels and no segmentation masks. The 10,015 masks present in the folder therefore come from an undocumented source. Rather than score the model against an unattributable mask, HAM10000 was excluded before any inference was run. The previously reported 79.15% Dice is retracted (item B, Section 9) and must not be cited anywhere.

Dataset-level independence is not established for any external cohort. The project's own Phase 32 audit concluded that no prospective clean external cohort is eligible. The 884-case study in Section 6 is a retrospective stress test: it measures how the frozen model behaves on data it was not trained on, and it does not establish generalisation because the cohorts were not held out prospectively and were previously inspected.

8. Statistical Limitations and Comparability

- No confirmatory significance testing is available. Phase 31 recorded zero new p-values, zero new effect sizes and zero eligible confirmatory procedures. No p-value appears anywhere in this report.
- VAL808 (n = 808) was consumed for model selection and early stopping, so all VAL808 numbers are development estimates.
- ROUTER_HOLDOUT (n = 875) was evaluated once and carries a disclosed post-first-open source-freeze deviation. It is not an independent external cohort.
- The Phase 21 six-model comparison and the BARR results come from different runs. Cross-run comparison is descriptive only; no controlled head-to-head between BARR and the baselines exists.
- Per-domain CIs for BUSI and KVASIR include zero, so those domains show no measurable difference.
- The external study is retrospective, not prospective. All three cohorts were previously inspected while building manifests and an evidence audit, and 8 hand-picked single cases had already been run under a different resize convention. No threshold, checkpoint or seed was chosen on external performance, but the cohorts are not a clean holdout.
- External uncertainty is a percentile bootstrap over clusters, described as such. It is not a confidence interval in a confirmatory sense, and no multiplicity correction is applied because no family of hypotheses is tested.
- Surface metrics on external data are in pixels. No cohort ships physical pixel spacing, so HD95, ASSD and NSD cannot be converted to millimetres and cannot be compared across cohorts.
- The Stones cohort is n = 20 with 6 positive cases. It carries no interval estimate; point estimates and counts are reported and no subgroup comparison is made.
- The BARR router threshold was calibrated on the internal development population. Its external selection fraction is an outcome of that fixed threshold, not a controlled property of the cohort.
- TEST (n = 1,683) remains locked and was not accessed for any value in this report.

9. Corrections and Retractions Relative to the October 30 Build

Items A–H below are the corrections applied to this build. A–D and H change what the report asserts about external data; E–G change how those assertions are framed and bounded.

#	Statement in the previous build	Status	Correction
A	External section presented an n = 1 per-dataset pilot, Table 6	SUPERSEDED	Replaced by the 884-case retrospective stress test in Section 6, produced by a design frozen before any external prediction existed.
B	HAM10000 Dice 79.15%	RETRACTED	HAM10000 excluded: the folder is the ISIC-2018 Task 3 classification release with no documented masks. No HAM10000 inference was run.
C	Figure 5 caption: "No BARR decision is shown because none was executed"	FALSE IN THIS BUILD	BARR has now been executed end-to-end on all 884 cases from verified artefacts. Figures 5–7 show the BARR prediction, BARR error map and decision trace.

#	Statement in the previous build	Status	Correction
D	"BARR Sω caught / rejected" external cases: no decision record exists	SUPERSEDED	A decision record now exists for every external case: 83 accepted and 35 rejected on CVC, 47 and 9 on BrEaST, 9 and 1 on Stones. These are counts, not a safety claim.
E	"The external material is a pilot, not a validation"	REFINED	Still not a validation, and now stated with the correct reason: the study is a retrospective stress test with no prospective holdout.
F	No quantification of domain shift	FILLED	Section 6.5 quantifies the shift against the internal VAL808 per-domain anchors: −13.4 pp (CVC), −15.1 pp (BrEaST), −68.3 pp (Stones).
G	External numbers reported without a stated aggregation mode	CORRECTED	Every external number now states macro-over-cases aggregation, its interval scheme, and that surface metrics are in pixels.
H	"No external validation exists"	SUPERSEDED	External measurement now exists and is reported with its limits. The claim that no external *validation* exists is retained, because none does.

Table 13. Correction and retraction register for this build. External evidence trail in reports/audits/c1_external_stress_v1_0/; earlier trail in reports/audits/c1_external_evidence_v1_0/.

The register below is carried forward unchanged from the October 30 correction pass. Those retractions remain in force; nothing in this build reinstates any of them.

Statement in the previous build	Status	Correction
Regional BARR macro Dice 86.24%	UNSUPPORTED	No stored record contains this value. Replaced by 83.79% on VAL808 (3-seed mean) and 93.03% on ROUTER_HOLDOUT, each with its population label.
BARR parameters 4.80M	UNSUPPORTED	Correct value 4,837,420 (4.84M) from the Phase 34 record.
Deltas marked "p<0.001" / "p<0.01" / "Significant"	UNSUPPORTED	No p-value was ever computed. Removed; deltas are descriptive with 95% bootstrap CIs.
"Sω eliminates 1,725 harmed cases"	MISATTRIBUTED	1,725 = A1→A8 system-level reduction. Sω alone accounts for 412→390 = 22 cases.

Statement in the previous build	Status	Correction
"BARR lifts TINY from 53.21% to 85.57%"	UNSUPPORTED	No BARR per-stratum measurement exists. 53.21% is the Global MedFiLM TINY stratum value. Removed.
"Independent external multi-center datasets"	OVERSTATED	Dataset-level independence is not established for any cohort; see Section 7.
BARR KUB Dice 85.57% and IoU 77.85%	UNSUPPORTED	These are the ROUTER_HOLDOUT KUB value and an unrecorded IoU. Removed from the VAL808 table.

Table 14. Retraction register carried forward from the October 30 build.

10. Trust Boundary Status

The C1 launch release is frozen at RC3 (digest 54d1273a…3549f) and is unmodified by this report. The production trust root was found to contain agent-generated signing material, and the RC3 gate currently refuses its own release digest because it delegates identity verification to RC2. Consequently no scientific execution is authorised, no production grant has been minted, and no dataset has been opened through a production launch. The trust store is excluded from scientific use pending accountable human review. This report does not create a trust policy, sign any approval, or constitute an authorisation.

11. What the Evidence Does Support

Defensible summary

1. Global MedFiLM reaches 83.51% macro-domain Dice on VAL808 (n = 808) with 3.52M parameters, ranking first of six under a single shared protocol, and is not statistically distinguishable from DeepLabV3+.
2. Regional BARR preserves macro Dice relative to Global MedFiLM: +0.28 pp on VAL808 (three-seed mean) and +0.17 pp on the one-shot ROUTER_HOLDOUT, within a −0.5 pp non-inferiority margin. This is a preservation result, not an improvement claim.
3. On in-sample development data the router removes most harmful refinements and the post-safety filter removes a further 22 cases, at 24.3% of the unconditional refiner workload.
4. On 884 external cases the frozen system runs end-to-end without error, and BARR changes the per-case Dice by a median of 0.00 pp in every cohort, with a macro change of +0.32 pp (CVC), +0.03 pp (BrEaST) and +0.13 pp (Stones). Preservation transfers; improvement does not.
5. External performance is far below internal performance: -13.4 pp on CVC-ClinicDB, -15.1 pp on BrEaST-Lesions-USG and -68.3 pp on Stones relative to the internal VAL808 anchors. On the Stones cohort the model marked foreground on 14 of 14 stone-free radiographs.
6. No external *validation* exists. The 884-case study is a retrospective stress test, and its cohorts were previously inspected, so it bounds failure modes rather than estimating generalisation.